

MITHIL K GOWDA

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Professional Summary

Software Engineering graduate with experience in backend application development, scalable software design, REST API development, database-driven application development, and secure software engineering. Project Intern at ISRO–LEOS with hands-on experience developing enterprise backend applications using Python, Flask, REST APIs, and MySQL. First author of internationally recognized research focused on AI-driven software architectures and intelligent distributed systems. Seeking graduate opportunities in Software Engineering, Backend Development, and Enterprise Software Development.

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru

2022–2026

B.Tech. in Information Science and Engineering

CGPA: **8.80 / 10**

Technical Skills

Programming Languages: Python, Java, JavaScript, SQL, HTML, CSS, Bash

Backend Development: Flask, REST APIs, Backend Application Development, API Design, Secure Software Development

Frontend Development: HTML, CSS, JavaScript, Responsive Web Design

Databases: MySQL, MongoDB

Software Engineering: Object-Oriented Programming (OOP), Data Structures Algorithms (DSA), Software Architecture, SDLC, Design Patterns, Problem Solving

Security: Authentication, Authorization, OWASP Top 10, Security Architecture

Tools: Git, GitHub, VS Code, Postman, Kali Linux

Cloud: Google Cloud Platform (GCP)

Operating Systems: Linux, Windows

Professional Experience

Indian Space Research Organisation (ISRO) – Laboratory for Electro Optics Systems (LEOS), Bengaluru
2026

Project Intern – Secure Software Development & Data Analytics

- Designed and developed scalable backend applications using Python, Flask, REST APIs, and MySQL to support engineering workflows and enterprise software solutions.
- Developed RESTful APIs, implemented secure authentication workflows, and optimized database interactions for reliable backend services.
- Built automated data processing and visualization pipelines to improve engineering analytics and operational efficiency.
- Engineered modular and maintainable backend components emphasizing software quality, scalability, performance optimization, code reusability, and maintainability.

Software Engineering Projects & Research

A Multi-Agent AI Cyber Defense Framework Using HoneyBee Method

- First Author and Corresponding Author of an AI-driven software framework accepted for presentation at ICIICE 2026 (Dubai, UAE) and accepted as a book chapter in the edited volume *Metaheuristic Optimization for Social Good*.
- Designed and developed a scalable multi-agent software architecture integrating intelligent threat detection, adaptive decision-making, and distributed system coordination.

- Implemented machine learning models including XGBoost, CNN-LSTM, and Deep Q-Network (DQN) within a modular software architecture for autonomous distributed decision making.
- Evaluated the framework using CIC-IDS2017 datasets and simulated enterprise scenarios, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, and 0.992 ROC-AUC.

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

- Abstract accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Designed and developed a secure distributed software platform integrating Digital Carbon Passports, authentication, authorization, fraud detection, trust scoring, and transaction management.
- Developed the Secure Trade Authorization and Verification Pipeline (STAVP) utilizing AES-256 encryption, SHA-256 hashing, authentication mechanisms, and secure transaction verification.

C³T-STAVP Cryptographic Framework (Ongoing Research)

- Abstract accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Researching Zero-Knowledge Proofs (ZKP), post-quantum cryptography, semantic obfuscation, secure software architectures, distributed systems, and resilient transaction validation mechanisms.
- Developed the Secure Trade Authorization and Verification Pipeline (STAVP) utilizing AES-256 encryption, SHA-256 hashing, secure authentication, and enterprise transaction validation.

Professional Certifications

- IBM Front-End Developer Specialization – IBM
- IBM IT Fundamentals for Cybersecurity Specialization – IBM
- Google Cloud Computing – NPTEL
- Ethical Hacker – Cisco Networking Academy
- Systems and Usable Security (Elite) – NPTEL
- Introduction to Cyber Security – IGNOU (SWAYAM)
- Introduction to Internet of Things (Silver Medal) – NPTEL
- Business Intelligence and Analytics – NPTEL

Achievements

- Selected as Project Intern at the Indian Space Research Organisation (ISRO – LEOS).
- First Author of internationally recognized research focused on AI-driven distributed systems, intelligent software architectures, and cybersecurity.
- Awarded the Reliance Foundation Undergraduate Scholarship (2026).
- Recipient of NPTEL Silver Medal in Introduction to Internet of Things.